

Introduction to Visual Studio Code

The Developer's Swiss Army Knife

What is VS Code?

Think of VS Code as a **well-equipped workshop**:

- **Tools** – Built-in coding features
- **Tool Storage** – Extensions for specialized tasks
- **Workbench** – Clean, organized workspace
- **Magnifying Glass** – Debugging & analysis
- **Reference Books** – Integrated documentation

Why Choose VS Code?

- **Free & Open Source** – No licensing fees, community-driven
- **Lightweight but Powerful** – Fast startup, rich features
- **Universal Language Support** – Works with 50+ languages
- **Extensible** – 40,000+ extensions to customize your workflow
- **Integrated Development** – Git, terminal, debugger built-in
- **Cross-Platform** – Same experience on Windows, macOS, and Linux

VS Code vs. Other Editors

Feature	Basic Text Editor	VS Code	Heavy IDE
Speed			
Features	Minimal	Balanced	Packed
Learning Curve	Easy	Moderate	Steep
Cost	Free	Free	Often Paid

What Can VS Code Do?

- **Smart Editing** – Syntax highlighting, IntelliSense, snippets
- **Code Navigation** – Go to definition, find references, symbol search
- **Debugging** – Breakpoints, step-through, variable inspection
- **Version Control** – Git integration, visual diffs, commit management
- **Project Management** – Multi-folder workspaces, file explorer, integrated terminal

- **Customization** – Themes, keyboard shortcuts, layouts, custom extensions

It is easy to make a VSCode extension once you master JavaScript programming.

Installation

1. Download from code.visualstudio.com
2. Verify the CLI tool:

```
code --version
```

Projects in VS Code

- In VSCode, a **folder** is a project.
- Create a `.vscode` folder in your project for custom settings

Common Commands

Open Folder

File → Open Folder (Ctrl+K Ctrl+O)

New File

File → New File (Ctrl+N)

Save

File → Save (Ctrl+S)

Command Palette

- **Shortcut:** `Ctrl+Shift+P` / `Cmd+Shift+P`
- One search bar for all VS Code features
- Fuzzy search, keyboard-first productivity

IntelliSense

Just like predictive text on your phone:

```
pr → print()
```

```
console. → log() | error() | warn()
```

Integrated Terminal

Shortcut: `Ctrl+`` / `Cmd+``

```
python my_script.py      # Run Python
pip install requests     # Install packages
git add .                # Git commands
```

Example: Python Project

1. Open Folder: `my_first_project`

2. Create File: `hello.py`

3. Write Code:

```
print("Hello, VS Code!")  
name = input("Your name: ")  
print(f"Nice to meet you, {name}!")
```

4. Run: Right-click → *Run Python File in Terminal*

Debugging Basics

- **Breakpoints** – Stop at specific lines
- **Step Through** – F10 (over), F11 (into), F5 (continue)
- **Inspect Variables** – Hover to see values, use Watch panel

We must install Python-related extensions to use programming features.

Customizing VS Code

Themes

File → Preferences → Color Theme

Keyboard Shortcuts

File → Preferences → Keyboard Shortcuts

Settings

File → Preferences → Settings (Ctrl+,)

We can edit JSON files to change the settings.

Local vs. Global Settings

- **Project-specific:** `.vscode/settings.json` in your folder
- **Global:**
 - Mac: `~/Library/Application Support/Code/User`
 - Windows: `%APPDATA%\Code\User`

Git Integration

- **Stage Changes:** Click  next to files
- **Commit:** Add a message, click ✓
- **Push/Pull:** Sync with GitHub or other repos

Search Shortcuts

- Find in File: `Ctrl+F` / `Cmd+F`
- Find in All Files: `Ctrl+Shift+F` / `Cmd+Shift+F`
- Quick Open File: `Ctrl+P` / `Cmd+P`

Essential Shortcuts

Action	Windows/Linux	macOS
Command Palette	Ctrl+Shift+P	Cmd+Shift+P
Quick Open	Ctrl+P	Cmd+P
Toggle Terminal	Ctrl+`	Cmd+`
Comment Code	Ctrl+/*	Cmd+/*
Format Document	Shift+Alt+F	Shift+Opt+F